B.Sc. Engg. SWE 3rd Semester

31 october 2024 (Afternoon)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2023-2024
FULL MARKS: 75

CSE 4309: Theory of Computing

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) State the differences between a Deterministic Finite Automaton (DFA) and Nondeterministic Finite Automaton (NFA). Explain δ for DFA, NFA, and epsilon NFA (ϵ-NFA). 5 (CO1) (PO1)

   b) Design a DFA for accepting a set of strings over input alphabet $\sum = \{a, b\}^*$ for the language $L = \{a^2bwa^2 | w \in \{a, b\}^*\}$. 8 (CO2) (PO1)

   c) Finite automata are good models for computer with an extremely limited amount of memory. The controller for an automatic door is one such example. Show the state diagram of this finite automata with the required states and input conditions. 6 (CO2) (PO1)

   d) The classic game Pac-Man requires the player to navigate through a maze, eating pellets, and avoiding the ghosts who chase him through the maze. Occasionally, Pac-Man can turn the tables on his pursuers by eating a power pellet, which temporarily grants him the power to eat the ghosts. When this occurs, the ghosts behavior changes and instead of chasing Pac-Man, they try to avoid him. The ghosts in Pac-Man have four behaviors: randomly wander the maze, chase Pac-Man when he is within line of sight, flee Pac-Man after Pac-Man has consumed a power pellet, and return to the central base to regenerate. 6 (CO2) (PO2)
   Draw the state diagram of a finite automata that emulates the behavior of the ghosts in Pac-Man. Show the states that the ghosts might be in at any given moment and also what inputs they take to make a transition from one state to another state.

2. a) What are the applications of regular expressions and finite automata? 5 (CO1) (PO1)

   b) Assume the password policy for creating your student account is set as follows: 6 (CO2) (PO2)
   - Password should start with atleast one symbols such as !, #, $, @, etc.
   - Password should contain atleast one upper case letter or atleast one digit as substring.
   - Password can contain zero or more number of lower case letters.

   Derive the regular expression that generates the passwords based on the policies mentioned above and design a Finite Automata that accepts the strings generated by the regular expression.

   c) You are given an NFA, $N_1 = (Q_1, \sum, \delta_1, q_1, F_1)$ that accepts the language A, and an NFA, $N_2 = (Q_2, \sum, \delta_2, q_2, F_2)$ that accepts the language B. Show that there exists an NFA, N that recognizes the language $A \circ B$. 6 (CO1) (PO1)

   d) Convert the following Regular Expressions to equivalent NFAs: 4 + 4 (CO5) (PO1)
   i. $(((00)^*(11)) \cup 01)^*$
   ii. $(0 \cup 10)^* 010 (0 \cup 1)^*$

3. a) Show that language L= $\{ww|w \in \{0,1\}^*\}$ is not regular using the Pumping lemma of regular languages. 8 (CO1) (PO1)

b) Minimize the DFA shown in Figure 1 using the Equivalence theorem. 7 (CO3) (PO2)

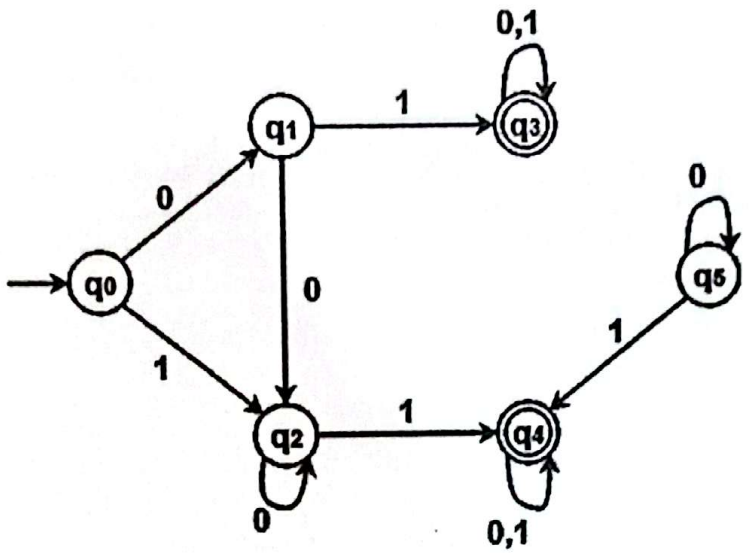

Figure 1: DFA State Diagram for Question 3.b

c) Convert the ϵ-NFA shown in Table 1 to an equivalent NFA by computing the ϵ-closure of each state. 10 (CO5) (PO1)

Table 1: Transition Table for Question 3.c

	a	b	ϵ
$\rightarrow q_0$	$\emptyset$	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_1\}$	$\emptyset$	$\{q_2\}$
$* q_2$	$\{q_1\}$	$\{q_0\}$	$\emptyset$

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24 April 2025

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Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION

DURATION: 3 HOURS

WINTER SEMESTER, 2023-2024

FULL MARKS: 150

CSE 4309: Theory of Computing

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Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) Define the terms alphabet, language, and grammar in the context of formal language theory. Provide examples for each term. — 7 (CO1) (PO1)

b) Design a Deterministic Finite Automata (DFA) for accepting a set of strings over input alphabet $\{a, b\}$ for the language, $L = \{abwa | w \in \{a, b\}^*\}$ where, w is the substring containing alphabets over any number of a and b ranging from zero to infinity. — 10 (CO2) (PO3)

c) Develop a grammar checker that checks and suggests corrections based on the grammatical context of an English sentence. Design a Finite Automata to represent the working of this grammar checker following the english grammar rules. — 8 (CO2) (PO3)

2. a) What are the closure properties of Regular Languages? Show that language L= $\{ 0^n 1^n \mid n \geq 1\}$ is not regular using Pumping lemma of Regular Languages. — 2 + 5 (CO1) (PO1)

b) If $L = N(P_N)$ for some Pushdown Automata (PDA) $P_N = (Q, \sum, \Gamma, \delta_N, q_0, Z_0)$, then prove that there is a PDA, P_F such that $L = L(P_F)$. — 8 (CO2) (PO3)

c) Convert the following Nondeterministic Finite Automata (NFA) shown in Table 1 to an equivalent DFA and informally describe the language it accepts. — 10 (CO5) (PO1)

Table 1: Transition table for Question 2.c

	0	1
$\rightarrow p$	$\{p, q\}$	$\{p\}$
q	$\{r, s\}$	$\{t\}$
r	$\{p, r\}$	$\{t\}$
$* s$	$\emptyset$	$\emptyset$
$* t$	$\emptyset$	$\emptyset$

3. a) Give a comparison among Finite Automata, Pushdown Automata, and Turing Machine. — 5 (CO1) (PO1)

b) Construct a PDA equivalent to the following Context-Free Grammar (CFG) and simulate it on the input string "$abbcbba$": — 8 (CO5) (PO1)

$$S \rightarrow aSa$$
$$S \rightarrow bSb$$
$$S \rightarrow c$$

c) What is Chomsky Normal Form (CNF)? Convert the following grammar into CNF: 12 (CO2) (PO3)

$$S \rightarrow ASB \mid \epsilon$$
$$A \rightarrow aAS \mid a$$
$$B \rightarrow SbS \mid A \mid bb$$
$$D \rightarrow \epsilon$$

4. a) What do you understand by ambiguous grammar? How can you remove ambiguity from a grammar? 5 (CO1) (PO1)

b) Design a PDA which accepts the if/else errors by empty stack. Show the moves for the input string, $w = iiieeee$. 10 (CO2) (PO3)

c) Design a PDA for the language L={$a^n b^m c^n d^m$| n,m ≥ 1} with transition diagram. 10 (CO2) (PO3)

5. a) What do you understand by Decidability, Undecidability and Intractability. 5 (CO1) (PO1)

b) Alan Turing proved that the Halting Problem is undecidable. Outline the key idea behind Turing's proof. 10 (CO1) (PO1)

c) Suppose you have a program $HALT(P, I)$ that correctly determines whether any given program P halts on input I. Explain how you could use this to construct a paradox, proving that $HALT$ cannot exist. 10 (CO1) (PO1)

6. a) Why are Turing Machines deterministic? Explain the tuples of a Turing Machine. 3 + 3 (CO1) (PO1)

b) Explain the components of a Turing Machine and their roles in the functioning of the machine. 7 (CO1) (PO1)

c) Design a Turing Machine for the language L={$a^n b^m a^{n+m}$| n,m ≥ 1} with transition diagram and show the tape movements for the string, $w = aabbbaaaaa$. 12 (CO2) (PO3)

aabbaaaa